



Social Networking Site Using MERN

PROJECT REPORT

SUBMITTED IN PARTIAL FULFILLMENT OF THE
REQUIREMENTS FOR THE AWARD
OF
DEGREE OF BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE
ENGINEERING



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A

Synopsis on

Social Networking Site Using MERN Stack

in partial fulfillment of the requirement for the degree
of

Bachelor of Technology

In

COMPUTER SCIENCE AND ENGINEERING

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Certificate

This is certified that this practical entitled "**Networking site**" is submitted by **Jaid Ahmed**, Roll No. 22069, student of Computer Science & Engineering Department, **Dronacharya College of Engineering**, Gurgaon in the partial fulfilment of the requirement for the award of Bachelors of Technology (Computer Science & Engineering) Degree of MDU, Rohtak, is a record of student's own study carried under my supervision & guidance.

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Introduction

The project is about Social networking site called “College Booth”.

This website is particularly designed for the college students, teachers , college staff and for upcoming students to resolve their queries about college with college students and with alumni also.

In this website they can chat with their college friends , post , share and comment on picture . In this web application college students can create their community for college clubs and society and interact with the interested one.

This website also helps the upcoming students or students coming for admission to resolve all their doubts and questions regarding with college students , alumni and with teacher , this will help students interaction among themselves.

For accessing this website particularly a college email-id is needed through which a user can create profile of his/her and enjoy the benefits of this site.

This web application is a best example of web based applications like facebook , instagram , twitter.

Technologies Used:

For UserInterface i.e Loginpage , Dashboard and all the other ui features we have used technologies like HTML and React for creating the structure , CSS and Bootstrap for styling the application

Frontend- Html , CSS , Javascript , React Framework

Backend – Java , Spring boot

Database- Nosql (Cassandra)

OBJECTIVE:

- Provides the searching facilities based on various factors. Such as Users, Shares, Photos, Friends.
- Users, Videos capture and modification is not possible.
 - It tracks all the information of Videos, Posts, Photos ect
 - Manage the information of Videos
 - Shows the information and description of the Users, Shares
- All the fields such as Users, Shares, Friends are validated and does not take invalid values
- It generates the report on Users, Videos, Posts
 - Provide filter reports on Shares, Photos, Friends
- You can easily export PDF for the Users, Posts, Photos
- Application also provides excel export for Videos, Shares, Friends
- You can also export the report into csv format for Users, Videos, Friends
- To increase efficiency of managing the Users, Videos
 - It deals with monitoring the information and transactions of Photos.
 - Manage the information of Users
 - Editing, adding and updating of Records is improved which results in proper resource management of Users data.
 - Manage the information of Photos
 - Integration of all records of Friends.

Existing Systems

System Analysis

1) STUDY OF THE SYSTEM

This application can be mainly divided into two modules:

1. Server
2. Client

This project “ignousocialnetworking.com” is mainly depended on client/server model. The client requests the server and server responses by granting the clients request.

The proposed system should provide both of the above features along with the followed ones:

SERVER: The server should be able to perform the following features:

The first and foremost problem is to find the server. We should identify the program in the server which processes the client’s request.

Administrator Client who will be acting as a super user.

Creating of private room with the password facility to enable private chats with the users online. The server is always waiting for clients requests .The clients come and go down but the server remains the same.

CLIENT: The client should be able to perform the following features:

Should be able to send message to anybody in the room with clients unique chat name created in the server for chatting purpose.

Should be provided with the drawing tools like free hand, rectangle, oval, line and also sending text message over the room.

In all the network applications, we find two sort program where the first i.e., server sends the information and the second i.e., client receives the information.

2) INPUT AND OUTPUT:

The user has to provide to which server it has to connect. In this, the server name to be provided is local host.

The user has to provide the username and the password for proceeding with the server for chatting purpose.

- **IDENTIFICATION OF NEED:**

Have u ever thought that we can talk with people residing At far of places i.e., orkut around the world now we can do this, that's what technology is all about and you will be surprised that you can talk with people all over the globe just sitting at your PC, sounds interesting yes!, just at the click of a button u can communicate and share information between different users who are present on their terminals at the time you logged in. By Social Networking Site a user can share his/her picture with many peoples at a same time, and know what people think about this picture. and they also know that his/her friends like this picture or not
ignousocialnetworking.com is do these all the things.

- **Preliminary Investigation:**

WWW is called the World Wide Web. WWW supports many kinds of text, pictures, video and audio. WWW resources through a web browser which basically a program that runs on the internet.

There are two kinds of browsers 1) text only browsers and 2) graphical browsers. Graphical browsers like Netscape Navigator and Internet Explorer are popular. These

Problem Statement

Our Project “College Booth” is a Web based application which particularly solves the problem of interaction among college students , teachers and college staff virtually .

This application also helpful for the upcoming students and also for our alumni to discuss their queries with students , teachers and college staff members.

It solves the problem of students to get interact with each other in a virtual manner and discuss about college clubs , societies by creating community among themselves which also helps college to grow in terms of extra-curricular and co-curricular activities.

Through this also college administration talk to students immediately when required and also for students to ask their query among teachers and college administration at one place.

This application also help the newly admitted students to get familiar with their seniors and discuss their questions with them.

This web based application help students to be updated about what is happening in college.

Proposed Methodology

SYSTEM DESIGN

- **Logical design (DFD)**

In the DFD, the inputs (source), outputs (destination), Databases (data stores), procedures (Data Flow) and boundaries of the system are described that needs the user requirement. It specifies the user need at a level of detail that virtually determines the information flow into and out of the system and the required data resources.

- **Physical design (Database design)**

This activity deals with the design of the physical database. A key is to determine how the access paths are to be implemented. A physical path is derived from the logical path. The relationships existing among the entities like one-to-one, one-to-many, many-to-many are considered while designing the database. Relational structured database is used in this system.

- **Program Design**

In conjunction with database design is a decision on the programming language to be used and the flowcharting, coding and debugging procedure prior to conversion.

In this application, java is used as server side scripting while JavaScript and HTML are used as client side scripting, which are supported by the browsers like Mozilla Explorer.

☐ Database Design

The Database (collection of data) is at the core of any information system. The nature of data stored in the database as well as the organization of the data has wide implications on the performance of the system. A good database design should hold all the data, which may be required by the users of the system, but at the same time, avoid any harmful or unnecessary redundancy. The database should also be flexible so that any future change in requirements can be incorporated with minimum changes.

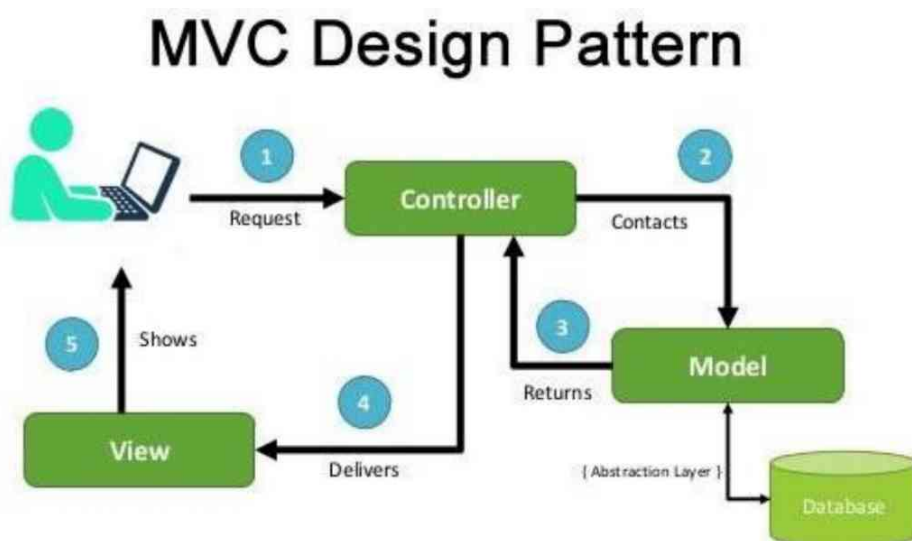
Implementation Methodology:

Model View Controller or MVC as it is popularly called, is a software design pattern for developing web applications. A Model View Controller pattern is made up of the

Following three parts:

- Model - The lowest level of the pattern which is responsible for maintaining data.
- View - This is responsible for displaying all or a portion of the data to the user.
- Controller - Software Code that controls the interactions between the Model and View.

MVC is popular as it isolates the application logic from the user interface layer and supports separation of concerns. Here the Controller receives all requests for the application and then works with the Model to prepare any data needed by the View. The View then uses the data prepared by the Controller to generate a final presentable response. The MVC abstraction can be graphically represented as follows.



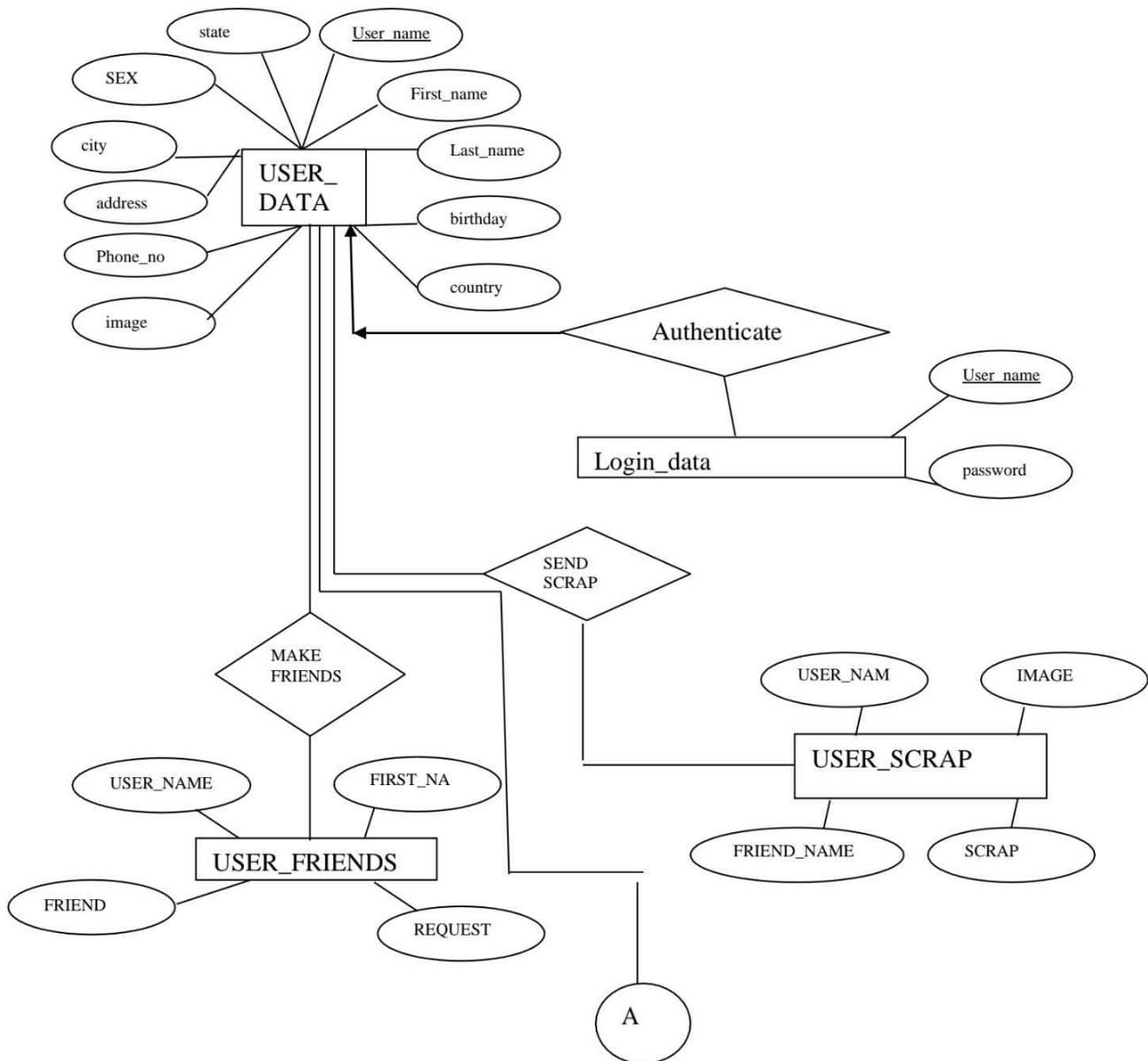
Advantages

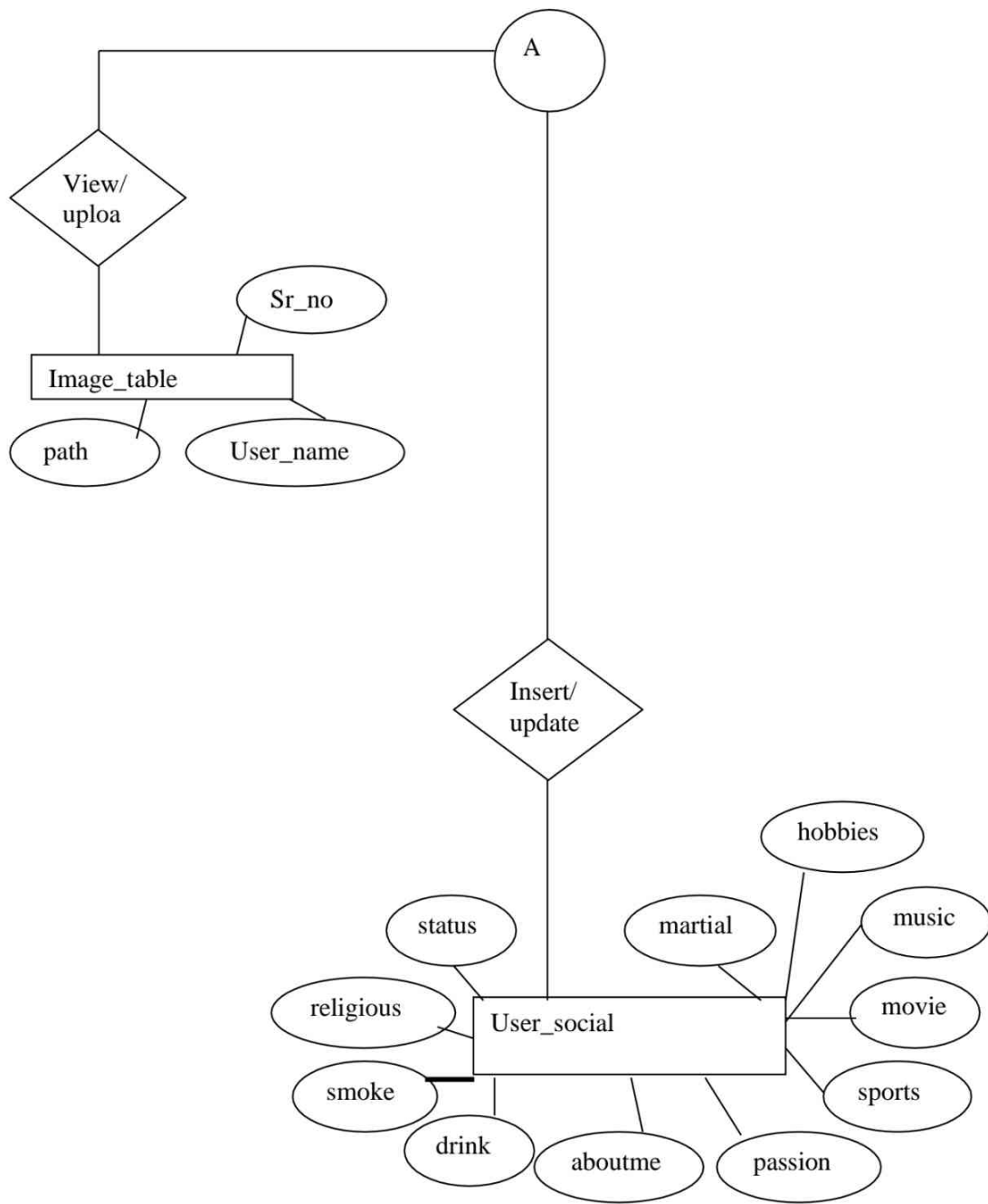
- Multiple developers can work simultaneously on the model, controller and views.
- MVC enables logical grouping of related actions on a controller together. The views for a specific model are also grouped together.
- Models can have multiple views.

Disadvantages

- The framework navigation can be complex because it introduces new layers of abstraction and requires users to adapt to the decomposition criteria of MVC.
- Knowledge on multiple technologies becomes the norm. Developers using MVC need to be skilled in multiple technologies.

ENTITY RELATIONSHIP DIAGRAM



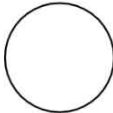
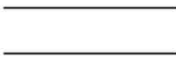


DATA FLOW DIAGRAM

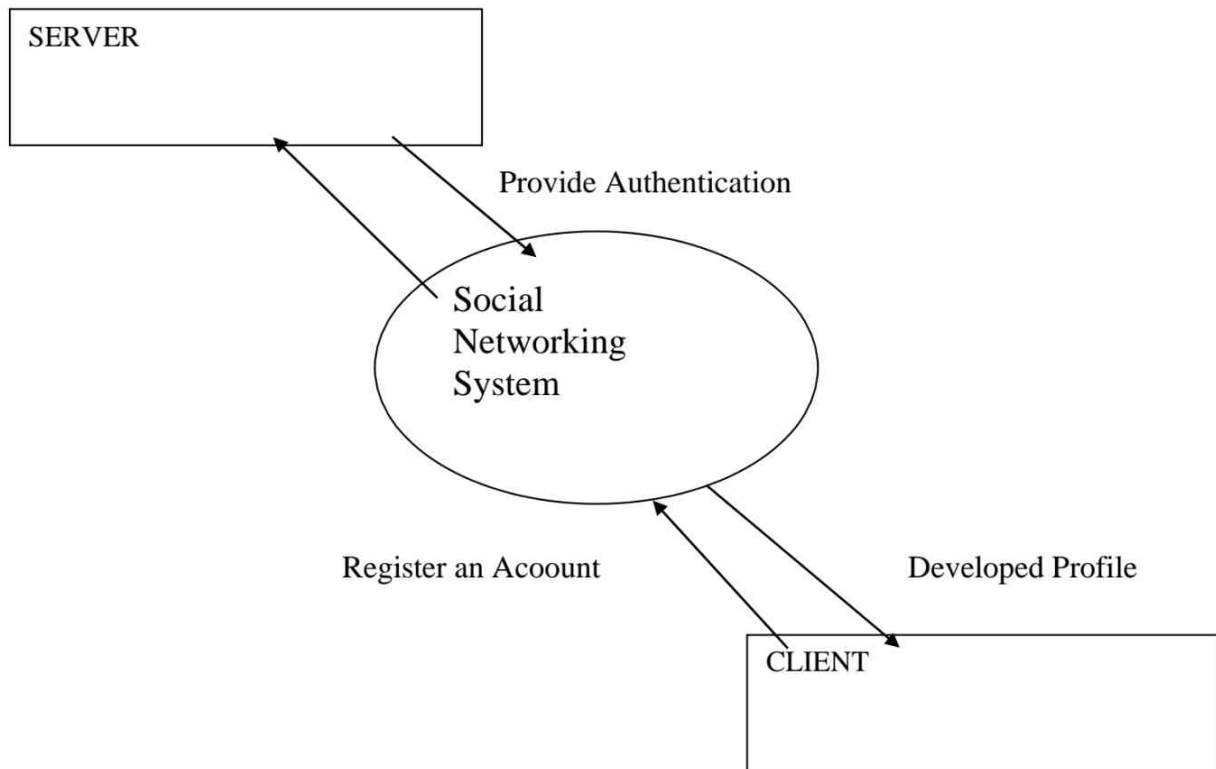
It is a pictorial representation of Business processes (functions/services/activities), along with the data flow.

Software process:-Guides how the software is being built.

Business process:-Specific to organization, e.g.:- In this project; issuing ticket, canceling ticket. In this focus is on what data flows and not how the data flows. When all the analysis is being made then we develop a diagram to depict the analysis, and following symbols are being used:-

S.NO.	SYMBOL	NAME	DESCRIPTION
1		Data Flow	Represents the connectivity between various processes.
2		Process	Performs some processing of input data.
3		External Entity	Defines source or destination of system data. The entity which receives or supplies information.
4		Data Store	Repository of data.

0 Levels DFD:



Feasibility Study

After doing the project Social Networking Site, study and analyzing all the existing or required functionalities of the system, the next task is to do the feasibility

study for the project. All projects are feasible - given unlimited resources and infinite time.

Feasibility study includes consideration of all the possible ways to provide a solution to

the given problem. The proposed solution should satisfy all the user requirements and

should be flexible enough so that future changes can be easily done based on the

future upcoming requirements.

A. Economical Feasibility

This is a very important aspect to be considered while developing a project.

We

decided the technology based on minimum possible cost factor.

- All hardware and software cost has to be borne by the organization.

Overall we have estimated that the benefits the organization is going to receive

from the proposed system will surely overcome the initial costs and the later on

running cost for system.

B. Technical Feasibility

This included the study of function, performance and constraints that may affect

the ability to achieve an acceptable system. For this feasibility study, we studied

complete functionality to be provided in the system, as described in the System

Requirement Specification (SRS), and checked if everything was possible using

different type of frontend and backend platformst.

C. Operational Feasibility

No doubt the proposed system is fully GUI based that is very user friendly and all

inputs to be taken all self-explanatory even to a layman. Besides, a proper training has

been conducted to let know the essence of the system to the users so that they feel

comfortable with new system. As far our study is concerned the clients are comfortable

and happy as the system has cut down their loads and doing.

Facilities Requirements

REQUIREMENT SPECIFICATION

Hardware Environment:

- Processor : Pentium IV, 600 MHz
- RAM : 256 MB
- Hard Disk : 40 GB
- Monitor : “SAMSUNG 15”

Software Environment:

- Operating System : Windows-XP
- Front End : HTML,CSS,Recat.js,Bootstrap
- Back End : Java , SpringBoot , NoSql Server(Cassandra)
- Development Tool : Visual Studio

Conclusion

Conclusion of the Project Social Web Application:

Our project is only a humble venture to satisfy the needs to manage their project work. Several user friendly coding have also adopted. This package shall prove to be a powerful package in satisfying all the requirements of the school. The objective of software planning is to provide a frame work that enables the manger to make reasonable estimates made within a limited time frame at the beginning of the software project and should be updated regularly as the project progresses.

At the end it is concluded that we have made effort on following points...

- A description of the background and context of the project and its relation to work
- already done in the area.
- Made statement of the aims and objectives of the project.
- The description of Purpose, Scope, and applicability
- We define the problem on which we are working in the project.
- We describe the requirement Specifications of the system and the actions that
- can be done on these things.
- We understand the problem domain and produce a model of the system, which
- describes operations that can be performed on the system.
- We included features and operations in detail, including screen layouts.
- We designed user interface and security issues related to system.
- Finally the system is implemented and tested according to test cases.

Future Scope of the Project:

In a nutshell, it can be summarized that the future scope of the project circles around maintaining information regarding

- We can add printer in future.
- We can give more advance software for Social Networking Site including more
- facilities
- We will host the platform on online servers to make it accessible worldwide
- Integrate multiple load balancers to distribute the loads of the system
- Create the master and slave database structure to reduce the overload of the
- database queries
- Implement the backup mechanism for taking backup of codebase and database
- on regular basis on different servers

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